Supporting Information

The quantitative currency of GPCR bias signalling fails its own algebraic identities

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This Supporting Information documents the complete computational output of the accompanying repository (`gpcr-field-audit`): every analysis script, the adjudication tables, the per-source meta-audit table, and the provenance of third-party data. Numbers are transcribed verbatim from `results/`; scripts are plain Python 3 (numpy/scipy/matplotlib/pandas). The fifteen scripts run in manuscript order under `run_all.py`, which archives each console output to `results/logs/` and writes a manifest recording package versions, exit codes and file hashes; `PAPER_MAP.md` maps every script to the panel or table it backs.

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1 Pipeline map: scripts and outputs

§	Script	Computes	Archived output
§3.1	code63	D2R binding-versus-function fracture (Fig. 1a, 2b)	results/logs/
§3.1	code64	D2R upstream rate ordering	results/logs/
§3.1	code68	D2R kinetic occupancy model (Fig. 2c)	results/logs/
§3.1	code69	D2R cross-arm spread versus time (Fig. 2a,c)	results/logs/
§3.2	code65	AT1R cross-arm affinity fracture (Fig. 1b, 2d)	results/logs/
§3.3	code66	μOR cell-level identity map (Fig. 1c, 2e)	results/logs/
§3.3	code80	μOR one-coordinate-per-arm model (Fig. 1c, 2f)	code80_*.csv, five files
§3.4	code67	Atlas meta-audit and pair deviations	results/logs/; atlas_pair_violations.csv under --write
§3.4	code71	Atlas identity pipeline; entry, group and paper tables	atlas_audit/*.csv, atlas_audit_summary.json
§3.5	code73	common-mode error, aggregation, jackknife	atlas_confidence_matrix.csv
§3.6	code77	constraint projection, fracture probability	atlas_repair_projection.csv, atlas_fracture_probability.csv
§4	code72	false-positive rate, power, controls	results/logs/
§3.4	code72b	fixed-cut independent-error null (entry and pair level)	results/logs/
SI §2	code47	P5 blind adjudication (208 cells)	Keshelava2018_P5/
SI §2	code47b	P5 erratum re-adjudication (195 cells)	Keshelava2018_P5_47b_erratum/

Rows are in manuscript order. Full script names, the reproduction command and the caveats (pandas grouping; which numbers are console-only) are in PAPER_MAP.md.

2 P5 adjudication (Keshelava 2018)

Dataset: per-cell Fura-2 peaks, 27 experiments × 433 cells × 7 ACh doses × 5 repetitions (Supplementary Data 3 of Keshelava et al., *Nat. Commun.* 9, 876 (2018)); 11 experiments passed the inclusion criteria. Frozen bootstrap seed 20260815, 2000 resamples, *AUC* with 95% CI.

Environment note. The tables below transcribe the archived run of 2026-08-16. Re-running the same scripts on the same data under the pinned environment of `requirements.txt` admits one more experiment to the timing chain (11 → 12), which shifts the cell counts (208 → 233 before the erratum, 195 → 218 after it) and moves the unit AUCs by at most 0.03 (robust arms: 0.13 and 0.23). Every adjudication clause reproduces verbatim: before the erratum P5-1 and P5-2 are intermediate and the overall verdict is intermediate; after the erratum P5-1 is falsified (2/4 units outside the [0.40, 0.60] band) and the overall verdict is falsified. The experiment-admission step is the numerically sensitive part; the verdicts are not.

2.1 Erratum adjudication (code47b; 195 cells, 13 timing exclusions)

Table 2: Unit table (verbatim): discrimination AUCs between adjacent dose steps, per statistic (τ = time above threshold; cnt = threshold crossings; pk = peak).

Statistic	pair	AUC	95% CI	n
τ	D1↔D2	0.629	(0.598, 0.662)	195
τ	D2↔D3	0.659	(0.625, 0.692)	195
τ	D5↔D6	0.568	(0.539, 0.599)	195
τ	D6↔D7	0.520	(0.487, 0.554)	195
cnt	D1↔D2	0.629	(0.598, 0.662)	195
cnt	D2↔D3	0.659	(0.626, 0.691)	195
cnt	D5↔D6	0.570	(0.550, 0.593)	195
cnt	D6↔D7	0.522	(0.501, 0.543)	195
pk	D1↔D2	0.731	(0.692, 0.770)	195
pk	D2↔D3	0.744	(0.716, 0.776)	195
pk	D5↔D6	0.551	(0.531, 0.571)	195
pk	D6↔D7	0.523	(0.509, 0.538)	195

Table 3: Clause verdicts against the frozen decision lines.

Clause	Frozen line	Verdict
P5-1	hit iff 4/4 τ-AUC in $[0.40, 0.60)$; falsified iff $\geq$2 units outside	**falsified** (2/4 units outside the band)
P5-2	hit iff 4/4 count-AUC $\geq$ 0.60; falsified iff 4/4 $<$ 0.60	intermediate (2/4 units at line)
P5-3	$\geq$3/4 units peak-AUC $\geq$ 0.60 → calibration pass	dataset-level weak-decoding limitation (only 2/4 units $\geq$ 0.60)
overall		**P5 falsified** (boundary predictor dead in this form)

Robustness arms (archived as `code47b_robust_arm.csv` in `Keshelava2018_P5_47b_erratum/`): verdicts are stable under $k = 2$ and $k = 4$ smoothing, under a desensitisation-corrected statistic, and within the single largest experiment (`experiment_11.dot`, $n = 29$). The original code47 adjudication (208 cells, pre-erratum) is archived in `Keshelava2018_P5/` for traceability; its overall verdict was "intermediate".

3 Atlas meta-audit (code67/71)

Per-source median absolute deviation from the algebraic identities, in dex, over all receptor/pathway groups in the BiasedSignalingAtlas community tables (Table 4). The full pair-level violation table (all receptor×pathway×ligand comparisons) is archived as `atlas_pair_violations.csv` (2.8 MB).

Table 4: Per-source-paper median and maximal identity deviation (dex).

DOI	n_{grp}	n_{pairs}	median	max	receptors
10.1111/bph.13854	2	4	1.322	2.209	OPRK_HUMAN
10.1124/jpet.116.232561	2	226	1.077	4.071	CNR2_MOUSE
10.1124/mol.115.101980	4	50	0.982	4.990	CNR1_HUMAN
10.1124/mol.113.088880	5	50	0.827	3.676	ADRB2_HUMAN
10.1074/jbc.M116.757559	7	13	0.682	1.966	CCR2/CCR5_HUMAN
10.1073/pnas.1312515110	3	3	0.676	1.357	CXCR4_HUMAN
10.1073/pnas.1908662116	4	28	0.531	1.496	OPRM_MOUSE
10.3389/fphar.2020.00188	1	1	0.529	0.529	OPRK_HUMAN
10.1016/j.bcp.2018.04.021	8	48	0.526	1.843	AGTR1_HUMAN
10.1073/pnas.2000712117	6	1240	0.452	4.569	OPRD/OPRK/OPRM_HUMAN
10.1007/s00018-013-1461-7	2	16	0.429	0.722	OPRD_HUMAN
10.1111/bph.14913	5	214	0.418	4.556	OPRD/OPRK/OPRM (H+M)
10.1124/mol.110.068106	2	57	0.400	1.796	DRD2_HUMAN
10.1016/j.bmc.2017.08.037	2	91	0.355	1.654	DRD2_HUMAN
10.1038/ncomms13958	8	288	0.326	4.054	CNR2_HUMAN/MOUSE
10.1074/jbc.M115.704122	1	15	0.287	0.610	NPSR1_HUMAN
10.1074/jbc.M112.408534	3	18	0.283	0.614	ACM2_HUMAN
10.1038/s41598-017-15258-z	2	13	0.270	0.664	OPRM_HUMAN
10.1124/mol.115.099192	2	42	0.267	0.737	CNR1_HUMAN
10.1016/j.phrs.2017.08.008	3	15	0.263	0.765	CNR1_HUMAN
10.1016/j.phrs.2018.02.032	5	225	0.262	1.275	APJ_HUMAN
10.1021/acs.jmedchem.8b00047	1	136	0.255	0.844	AA1R_HUMAN
10.1038/nchembio.2527	2	13	0.220	0.342	DRD2_HUMAN
10.3390/molecules24244542	4	351	0.210	0.847	OPRD/OPRM (H+M)
10.1124/mol.115.099150	3	27	0.205	1.822	OPRX_HUMAN
10.1002/pep2.24070	4	288	0.196	1.714	OPRD_MOUSE/OPRM_HUMAN
10.1038/s41467-018-03941-2	2	132	0.194	1.480	GLP1R_HUMAN
10.3390/ijms20102417	5	46	0.183	0.560	CCR1_HUMAN
10.1126/scisignal.aaz3140	5	157	0.178	1.452	OPRM_HUMAN/MOUSE

DOI	n_{grp}	n_{pairs}	median	max	receptors
10.1124/jpet.114.220293	2	110	0.178	0.652	DRD2_HUMAN
10.1016/j.lfs.2016.02.069	1	6	0.167	0.328	EDNRA_HUMAN
10.1016/j.bcp.2014.05.008	3	15	0.163	0.728	AA2BR_ HUMAN/MOUSE
10.1038/s41467-019-11875-6	30	5399	0.162	3.325	ADRB2/OPRD/OPRM (H+RAT)
10.1016/j.euroneuro.2018.12.013	3	13	0.160	0.329	OPRD_HUMAN
10.1021/acschemneuro.9b00429	5	20	0.157	0.316	CNR1_HUMAN
10.1038/s41598-017-01078-8	2	2	0.153	0.249	LSHR_HUMAN
10.1038/s41598-020-65636-3	7	73	0.148	1.809	ADRB1_HUMAN
10.1016/j.bcp.2015.11.013	2	30	0.147	1.300	AA1R_HUMAN
10.1124/mol.115.098848	6	562	0.138	3.363	OPRM_HUMAN
10.1124/mol.113.089649	2	930	0.137	0.739	OPRK_HUMAN
10.1161/ HYPERTENSIONAHA.114.05099	3	3	0.132	0.762	APJ_HUMAN
10.1021/jm500457x	2	49	0.120	0.417	DRD2_HUMAN
10.1126/scisignal.aar4309	4	12	0.119	0.986	OPRK_HUMAN
10.1007/s00210-014-1054-5	5	417	0.117	0.568	ADRB2_HUMAN
10.1021/acs.jmedchem.8b01912	4	18	0.116	0.265	FPR1/FPR2_HUMAN
10.1124/mol.116.103283	2	210	0.115	0.588	AA3R_HUMAN
10.1124/mol.116.106153	4	40	0.112	0.688	HRH3_HUMAN
10.1021/acs.jmedchem.7b00363	2	110	0.110	0.745	DRD2_HUMAN
10.1126/science.1232808	4	4	0.109	0.331	5HT1B/5HT2B_ HUMAN
10.1124/mol.110.067454	1	21	0.107	0.273	ADA1A_HUMAN
10.1124/mol.110.065664	2	30	0.102	0.335	GLP1R_HUMAN
10.1126/scisignal.aat1631	8	524	0.101	1.532	AGTR1_HUMAN
10.1021/acsptsci.0c00022	3	217	0.100	0.693	GLP1R_HUMAN
10.1074/jbc.M113.504381	2	49	0.096	0.250	OPRK_HUMAN
10.3389/fphar.2018.01202	3	45	0.083	0.454	CNR2_HUMAN
10.1021/acs.jmedchem.6b01857	2	141	0.081	0.340	DRD2_HUMAN
10.1021/acs.jmedchem.6b01875	2	240	0.080	0.803	DRD2_HUMAN
10.1038/s41401-018-0200-x	2	342	0.078	0.316	ADRB2_HUMAN
10.1074/jbc.M115.659250	8	17	0.074	0.823	GHSR_HUMAN
10.1016/j.bcp.2017.03.014	4	12	0.067	0.190	AA2BR_HUMAN
10.1016/j.bcp.2016.05.014	6	168	0.064	0.818	OPRM_ HUMAN/MOUSE
10.1016/j.neuropharm.2019.107718	4	36	0.063	0.177	OPRM_HUMAN
10.1021/acsptsci.9b00083	2	342	0.050	0.327	CALRL_HUMAN
10.1016/j.bcp.2017.03.018	4	147	0.042	0.304	GLP1R_HUMAN
10.1021/jm401318w	2	218	0.040	0.296	DRD2_HUMAN
10.1101/2020.11.09.375162	18	3608	0.001	0.364	CNR1/CNR2_HUMAN

4 MOR extended-model residuals (code80)

The Gillis arm-coordinate extended model is fitted to the MOR ligand series; per-(nanobody, ligand, channel) residuals with standard deviations, model-1 residuals and arm shifts are archived verbatim in `mor_code80_residuals.csv` (67 rows). Example rows: Nb33 / Morphine: $\log \tau$ residual $+0.031 \pm 0.102$, $\log R$ residual -0.021 ± 0.092; Nb33 / Oxycodone: $\log \tau$ $+0.192 \pm 0.146$.

5 Third-party data provenance

Shipped here: `keshelava2018/peak_values.csv` (Supplementary Data 3 of Keshelava et al. 2018, CC BY 4.0; authors' MATLAB script carries a legacy dose ladder — our usage follows the Methods ladder, documented in the manuscript Methods). Referenced but not re-hosted: BiasedSignalingAtlas community tables (code67/71/77); Gillis et al., *Sci. Signal.* **13**, eaaz3140 (2020) main Tables 1–3 and SI Table S1 ($\sim$98 MB, not mirrored); D2R/AT1R source studies cited in the manuscript (Klein Herenbrink et al. 2016; Wingler et al. 2020). Full notes in `data_external/PROVENANCE.md`.